

WaveLens Lite — Technical Design Document

Bone-Conduction Conversational Interpreter · Agora × Solana
Convo AI Hackathon — Đại học Bách Khoa Đà Nẵng 2026 · Live Agora Docs

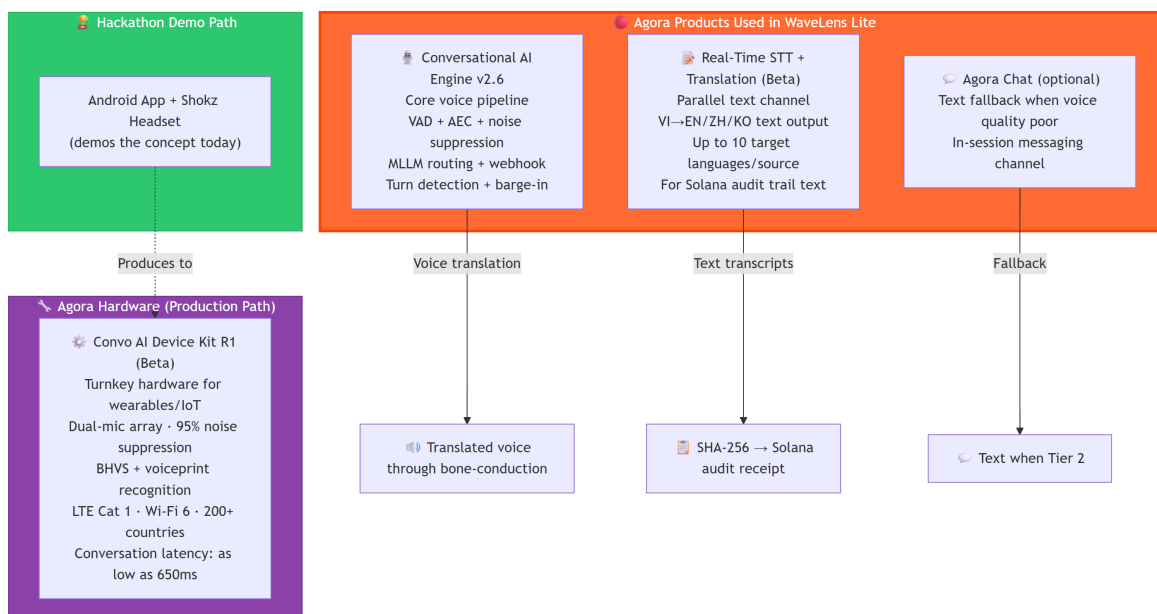
⚡ Hackathon Context

Chủ đề: Build Conversational AI bằng **Agora Conversation AI Engine**, tích hợp **Solana** để giải quyết một vấn đề thực tế trong cuộc sống.

Kickoff online	June 18, 2026
Build period	June 18 – 27, 2026 (9 days)
Demo Day offline	June 28, 2026 — ĐH Bách Khoa Đà Nẵng
Prizes	🏆 \$400 · 🥈 \$200 · 🥉 \$100 · + \$600 builder credits
Agora CAI free quota	300 minutes/month (shared with STT)
Agora CAI after free	\$0.10/minute
\$600 credits =	~6,000 minutes of Agora CAI testing

🏆 The Core Question: "Why Agora — and Which Agora Product?"

WaveLens Lite uses three Agora products simultaneously. This is the answer every judge needs to hear.



The three Agora products complement each other:

- **CAI Engine** delivers the real-time voice translation (the core demo moment)
- **Real-Time STT + Translation** produces the bilingual text for the Solana audit hash (a cleaner audit layer)
- **Convo AI Device Kit R1** is the production hardware path for industrial wearables — showing judges you understand the full product vision

0. Executive Summary

WaveLens Lite is a **real-time conversational interpreter** for Vietnamese workers in industrial environments — ports, ship engine rooms, maritime decks — communicating with English, Korean, or Chinese supervisors where phones are impractical and blocking your ears is dangerous.

Core stack (June 2026, live docs verified):

Layer	Technology	Role
Real-time voice transport	Agora SDRTN™	UDP audio, 80% packet loss tolerance
Voice AI orchestration	Agora CAI Engine v2.6	VAD, AEC, noise suppression, MLLM routing, webhooks
Text audit channel	Agora Real-Time STT + Translation (Beta)	Bilingual text (VI + EN) for Solana hash

Layer	Technology	Role
AI — Maritime sessions	<code>gpt-realtime-2</code> via Agora MLLM	Supports maritime glossary, bidirectional routing
AI — Coaching/General	<code>gpt-realtime-translate</code> via Agora MLLM	\$0.034/min, 70+ in → 13 out, no glossary
VI Input Transcription	<code>gpt-4o-transcribe</code> (hint: "vi")	For webhook transcript log
Offline STT	PhoWhisper medium (VinAI, local)	Tier 3 only — 844h Vietnamese training data
Backend	FastAPI + Redis	Session state, domain routing, Solana service
Audit trail	Solana Anchor — Receipt PDA	SHA-256 of bilingual transcript, immutable
Payment	Solana x402 — USDC	Per-session micropayment, 165M+ txn ecosystem
Hackathon hardware	Shokz OpenRun Pro 2 (SBC, IP55)	Demo output device
Production hardware	Agora Convo AI Device Kit R1 (Beta)	Integrated wearable solution with LTE + dual-mic

Latency targets: Voice translation: P50 ~670ms (gpt-realtime-2) · P90 <1,400ms

1. Problem Statement

1.1 Why Phones Fail in Industrial Environments

Environment	Noise	Hands-free	Water	Language Gap
Container port (cảng biển)	85–100 dB(A)	Yes (crane, forklift)	Rain, salt spray	VI ↔ EN, ZH

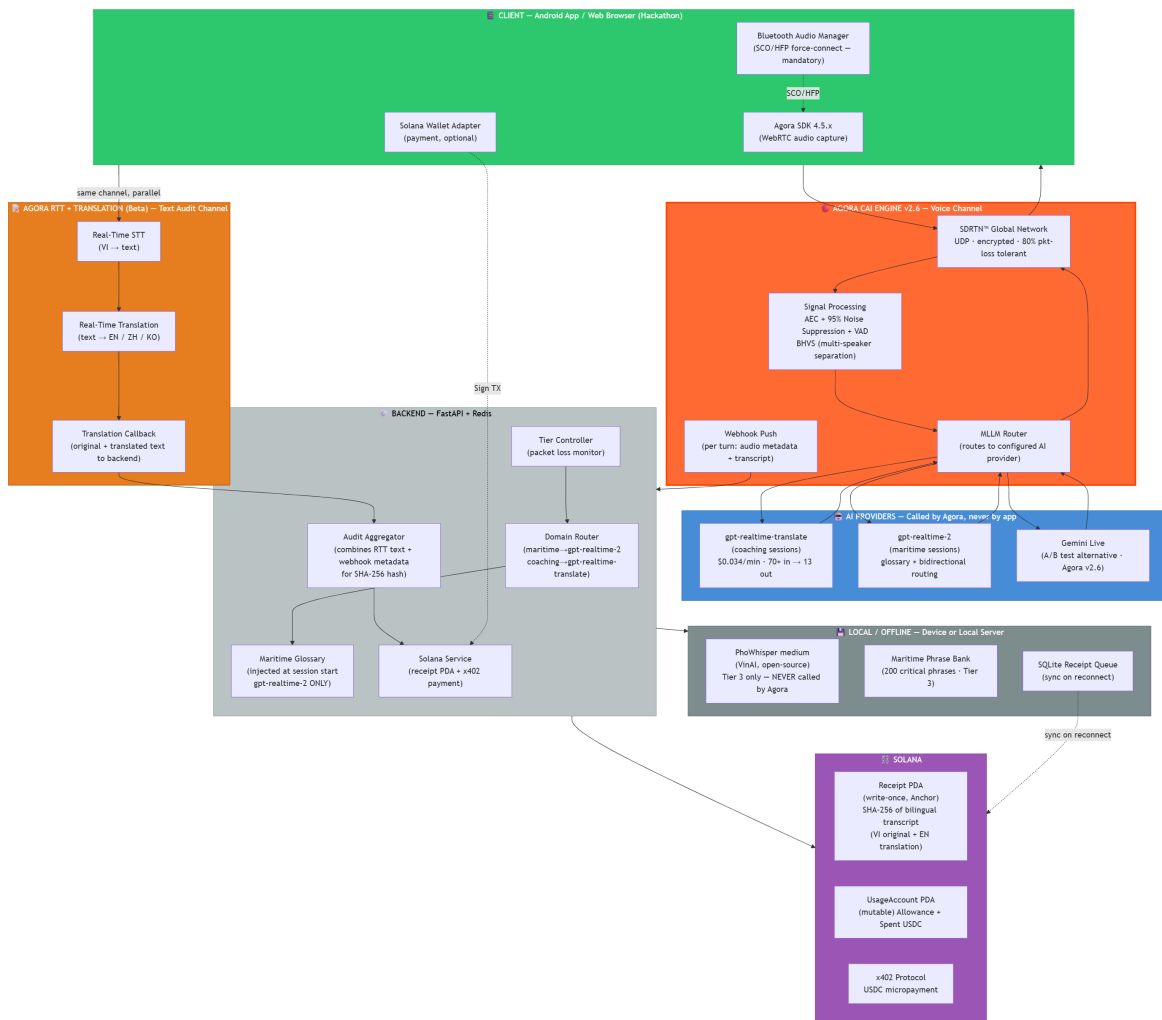
Environment	Noise	Hands-free	Water	Language Gap
Ship engine room	90–110 dB(A)	Yes (tools, valves)	Steam, bilge	VI ↔ EN, KO
Ship bridge / deck	70–95 dB(A)	Often	Rain, wave spray	VI ↔ EN, KO
Pool coaching	70 dB	Yes (swimming)	Full immersion	VI ↔ EN

1.2 Da Nang: The Right Context for This Problem

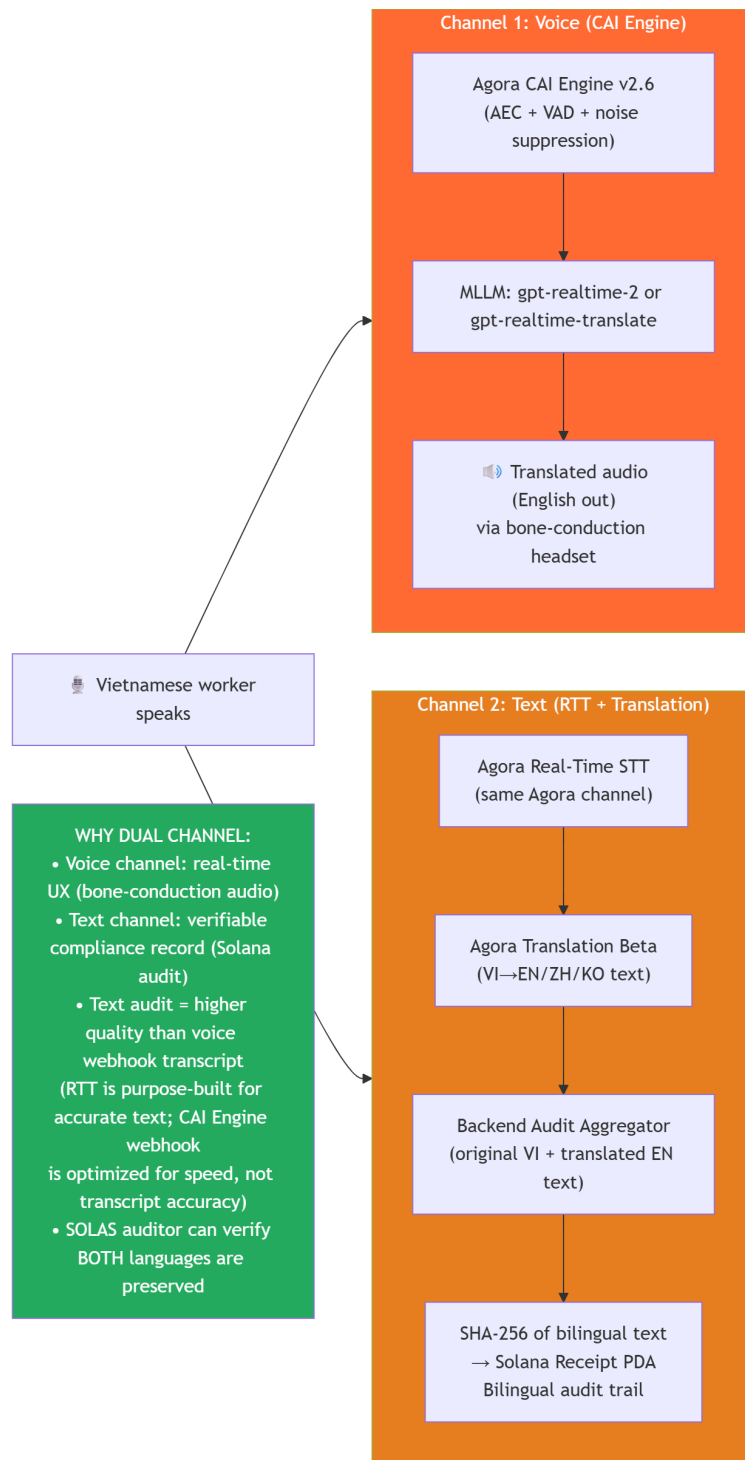
- **Da Nang Port 2026 target: 16.77 million tonnes** (9% YoY growth, official Jan 2026 announcement)
- Lien Chieu deep-sea port construction started Q1 2026 — first berths operational Q4 2028
- COSCO added new container route in 2025 — Korean and Chinese crews now regular at Da Nang
- East-West Economic Corridor connects Da Nang to Laos, Thailand, Myanmar — multilingual crews standard

2. System Architecture

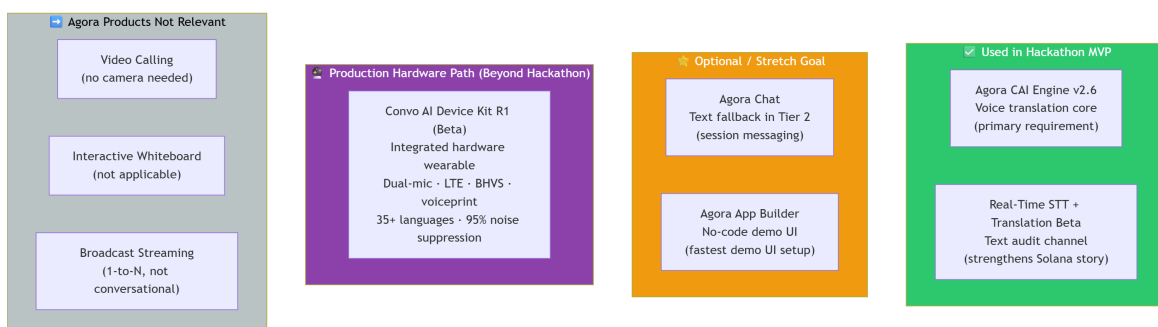
2.1 Full Component Diagram (Dual-Channel Design)



2.2 Why Dual Channel — Voice AND Text in Parallel



2.3 Full Agora Product Map for WaveLens



3. AI Model Selection

3.1 Two Models, Two Purposes (Confirmed from OpenAI Docs)

 gpt-realtime-2

✅ CAN DO

- Everything translate does
- Custom system prompts
→ glossary!
- Tool calling + reasoning
 - Multi-turn context retention
- /inject instructions (as user input)
- Bidirectional language routing

❌ COSTS MORE

- Token-based pricing
- Slightly higher latency
- More complex to configure

 gpt-realtime-translate

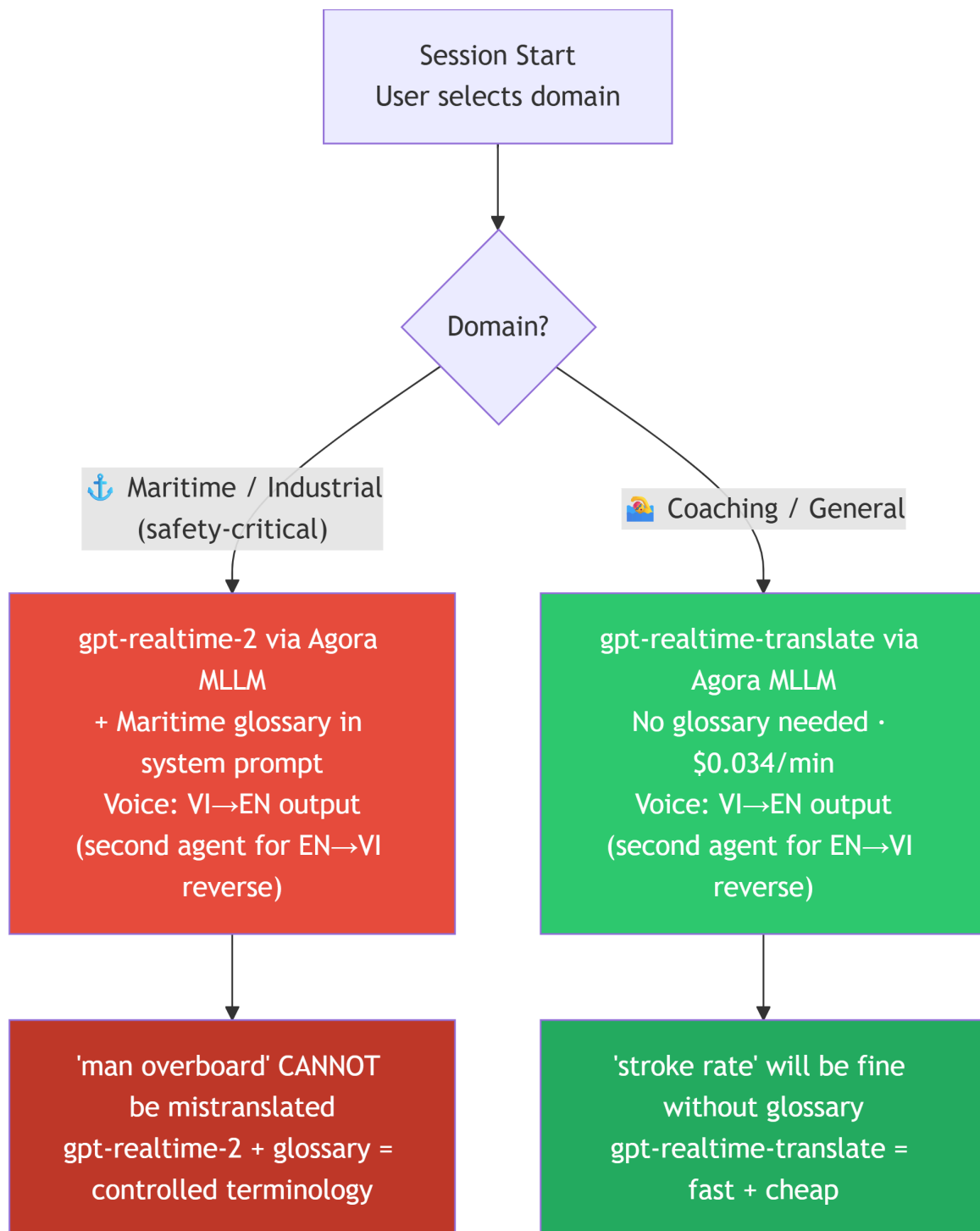
✅ CAN DO

- Speech→Speech (voice in, voice out)
- 70+ input / 13 output languages
- \$0.034/min (predictable)
- Professional interpreter pacing
- Dynamic voice adaptation
- Low latency (~200-400ms model)

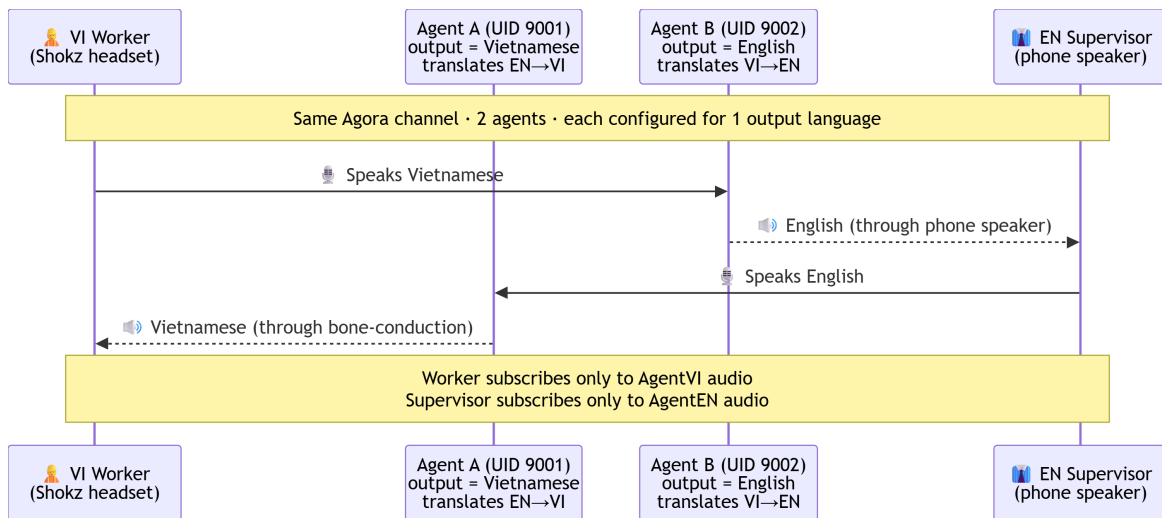
❌ CANNOT DO

- Custom system prompts
 - Glossary injection
- Pronunciation guides
 - Tool calling
- Conversation memory
- Agora /inject instructions

3.2 Domain Routing — Which Model for Which Session



3.3 Bidirectional Translation — Two-Agent Pattern (Confirmed)



3.4 Confirmed Output Languages (13 — All Use Cases Covered)

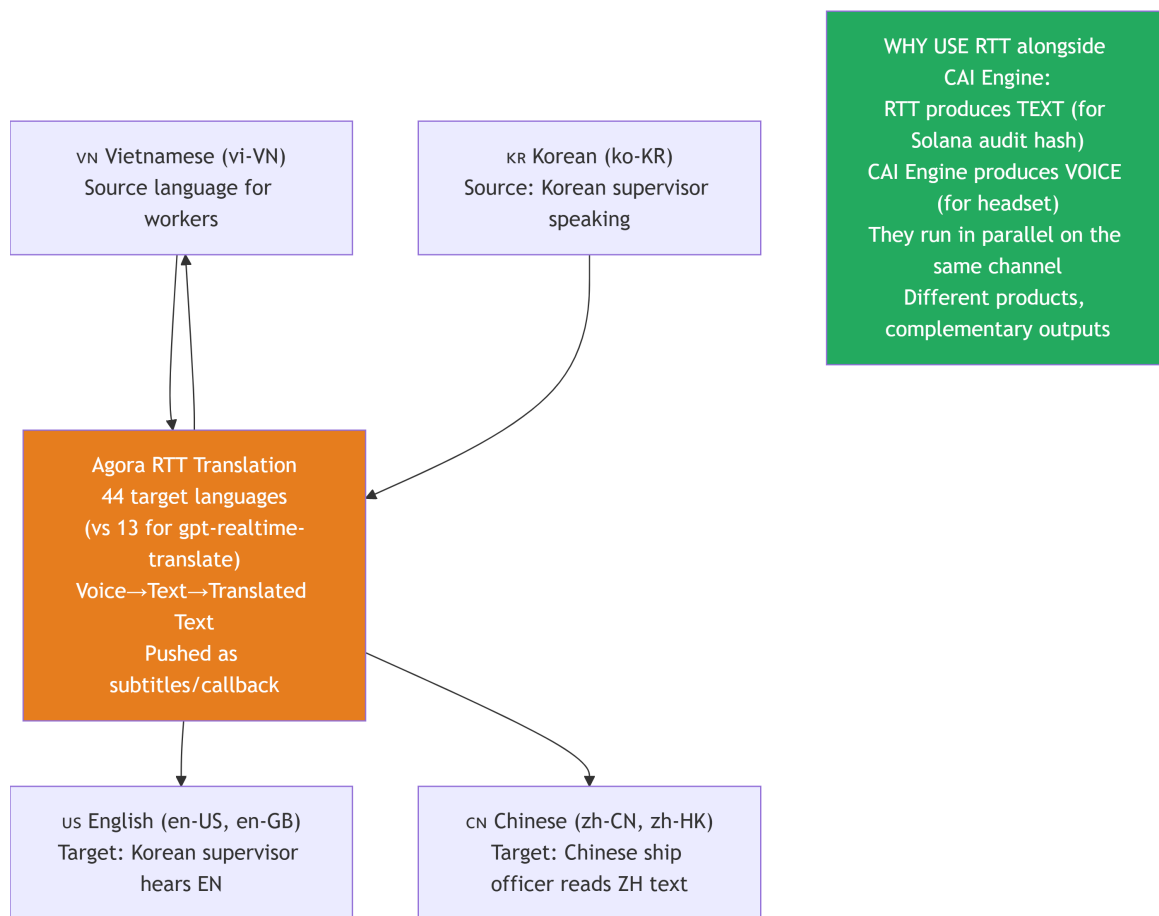
Confirmed by OpenAI Cookbook, May 7, 2026:

English · Chinese · Korean · Vietnamese · Spanish · Portuguese · French · Japanese · Russian · Hindi · Indonesian · Italian · German

All three WaveLens target output languages (EN, ZH, KO) are confirmed. **Vietnamese is ALSO an output language** — supervisors speaking EN/KO/ZH can be heard in Vietnamese through the worker's bone-conduction headset.

3.5 Agora RTT Translation — Supported Languages (Broader Than OpenAI)

The Agora Real-Time STT Translation (Beta) supports **44 target languages** from the official docs, including:

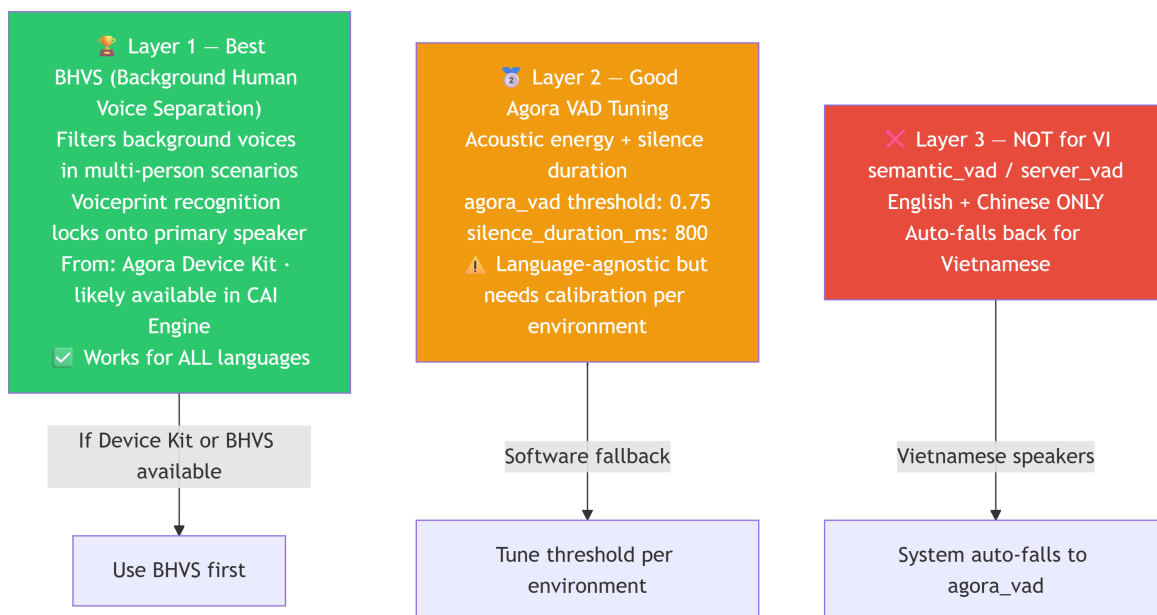


RTT Translation latency (from Agora docs): end-to-start under 1 second, average end-to-end under 3 seconds. This is too slow for real-time VOICE but perfectly fine for generating the text audit record.

4. Critical VAD Issue for Vietnamese

4.1 The Problem Hierarchy (Expanded)

Three layers of speaker isolation, from best to fallback:



4.2 BHVS and Voiceprint Recognition (From Agora Device Kit Docs)

Background Human Voice Separation (BHVS): Filters background voices in multi-person conversation scenarios — directly solving the problem of multiple port workers talking simultaneously while only one is intended for translation.

Voiceprint recognition: Locks onto the primary speaker in multi-person conversations. Once the worker's voice is registered, the system filters out other workers, crane operator chatter, and supervisors speaking in adjacent areas.

These algorithms run at the **Agora Convo AI Core level** — before audio even reaches the MLLM. For the hackathon:

- Configure `properties.audio.aec` and `properties.audio.ns` in the CAI Engine start call
- For BHVS-style isolation, use the voiceprint attention feature (available in some CAI Engine releases)

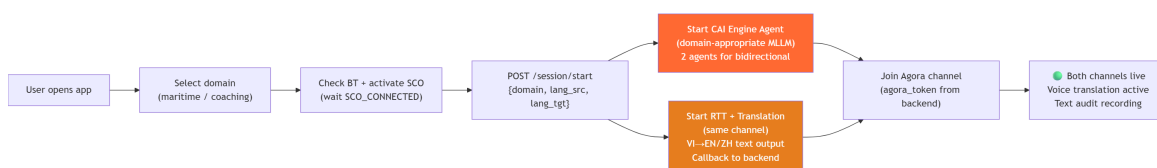
4.3 VAD Configuration by Environment

Environment	Recommended Mode	Threshold	Silence Duration	Primary Challenge
Container port (crane noise)	agora_vad	0.75	800ms	High ambient + multiple workers
Ship engine room	agora_vad	0.80	900ms	Loudest; machine vibration triggers

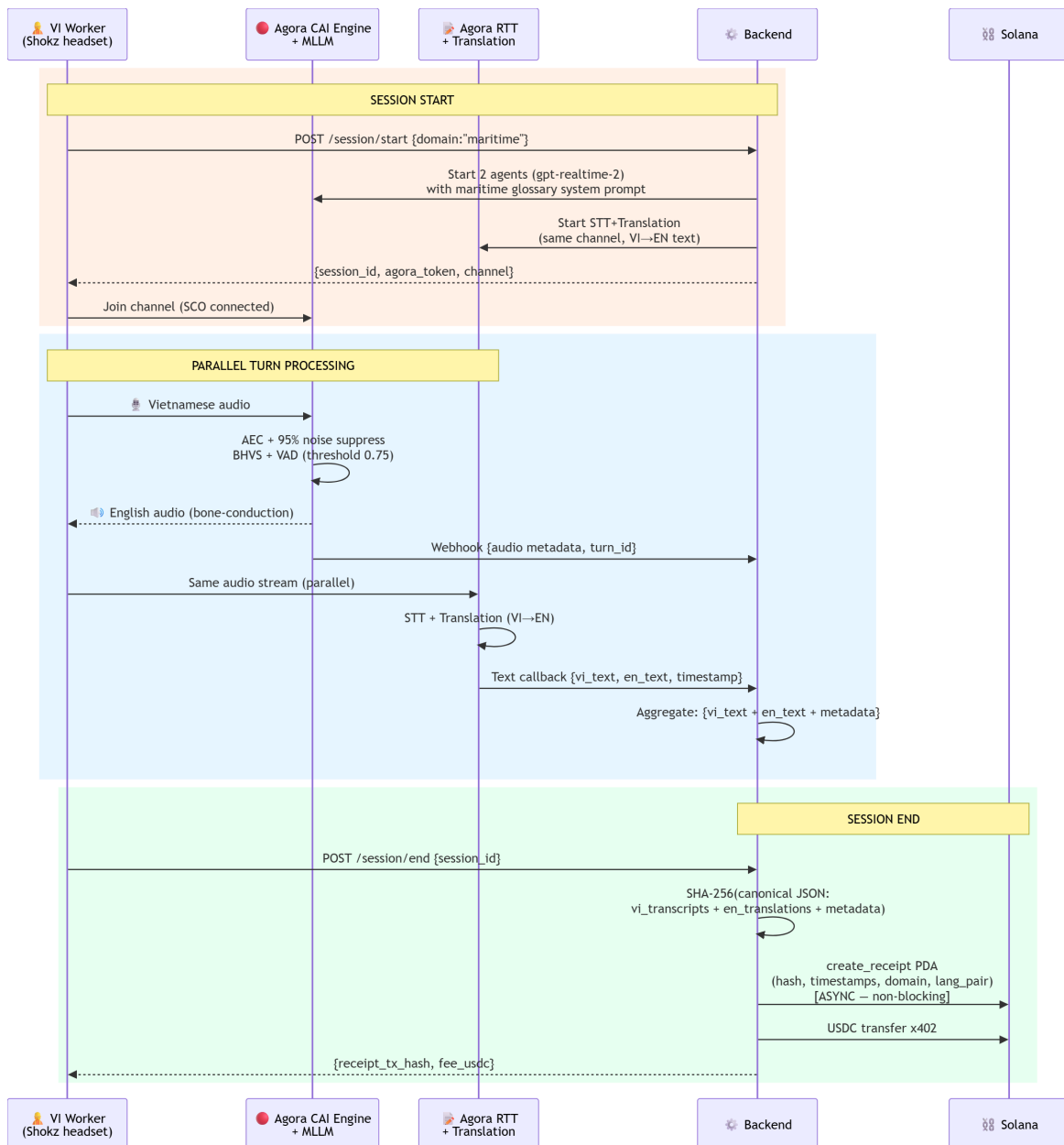
Environment	Recommended Mode	Threshold	Silence Duration	Primary Challenge
Ship bridge/deck	agora_vad	0.70	720ms	Wind gusts cause spikes
Poolside coaching	agora_vad	0.60	640ms	Quiet; default-like settings fine
EN/KO/ZH supervisor (quiet office)	semantic_vad	n/a	n/a	Use smart VAD when available

5. Session Flow

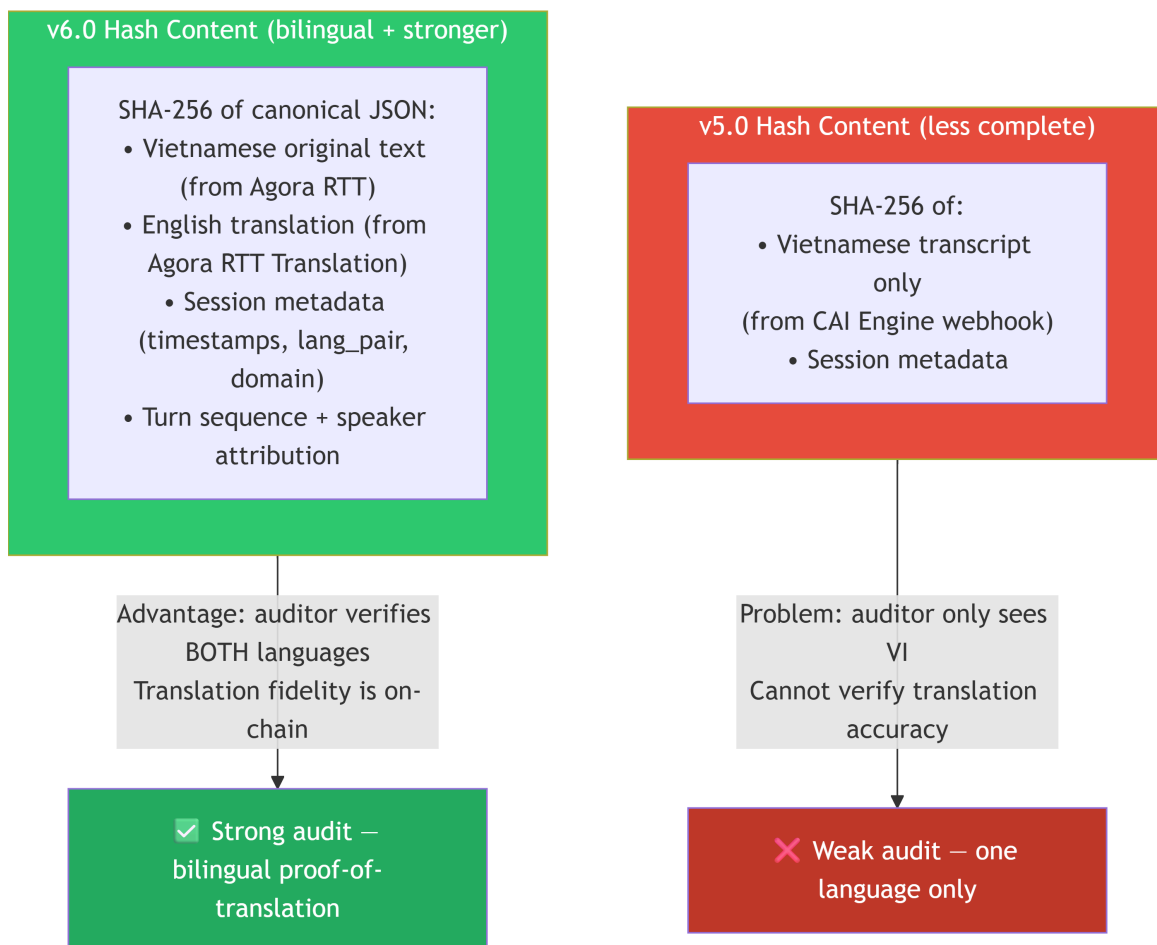
5.1 Dual-Channel Session Setup



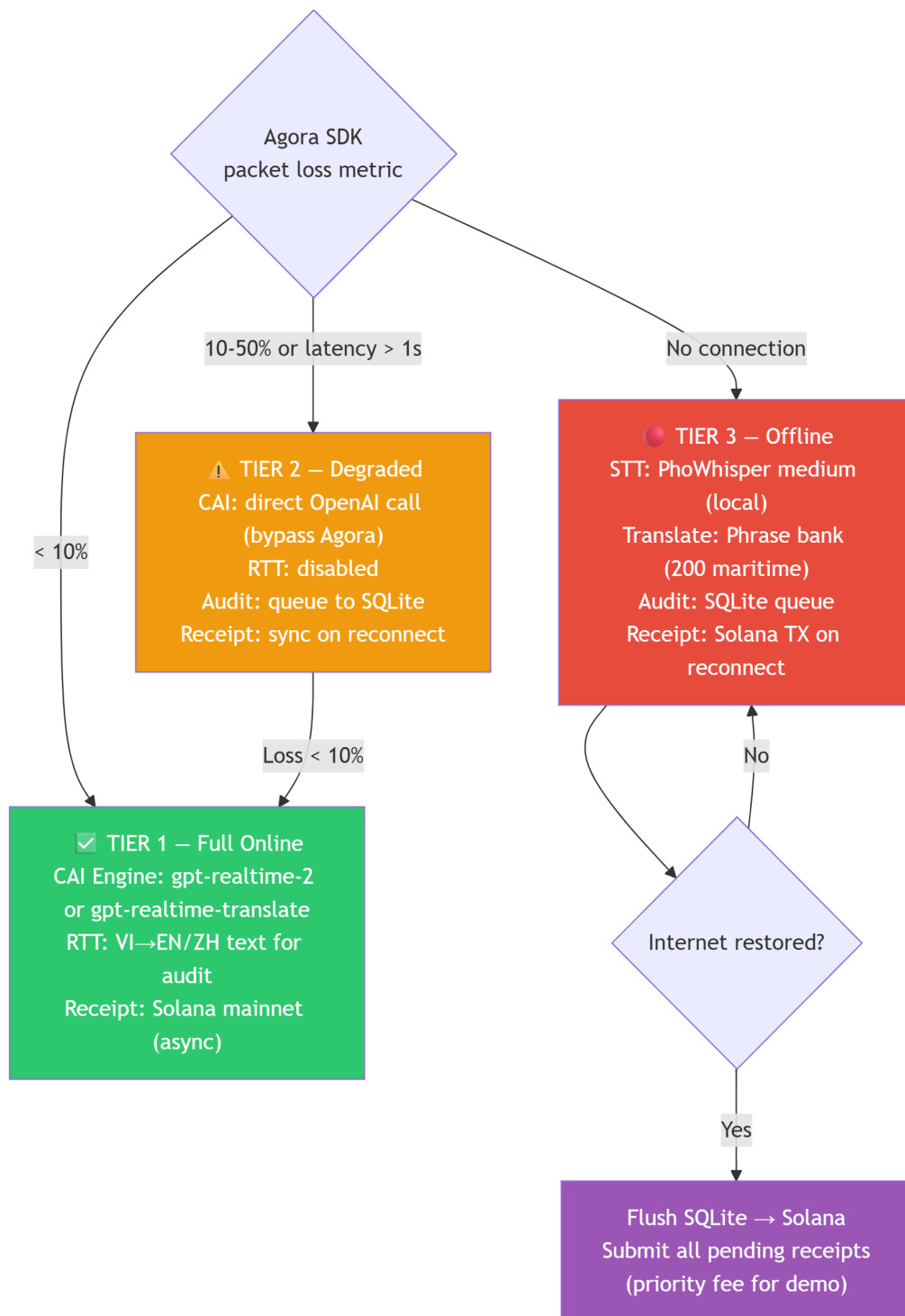
5.2 Full Conversational Turn — Sequence Diagram



5.3 What Goes into the Solana Hash



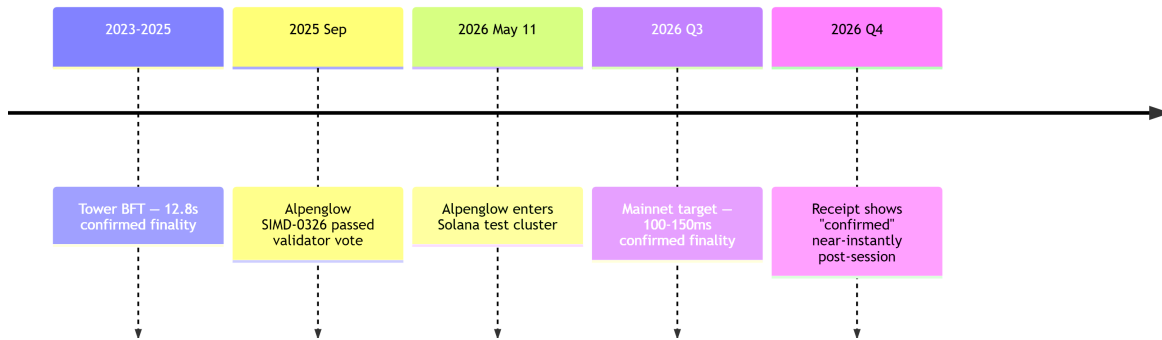
5.4 Offline Tier Selection (Unchanged — Confirmed Sound)



6. Solana Integration

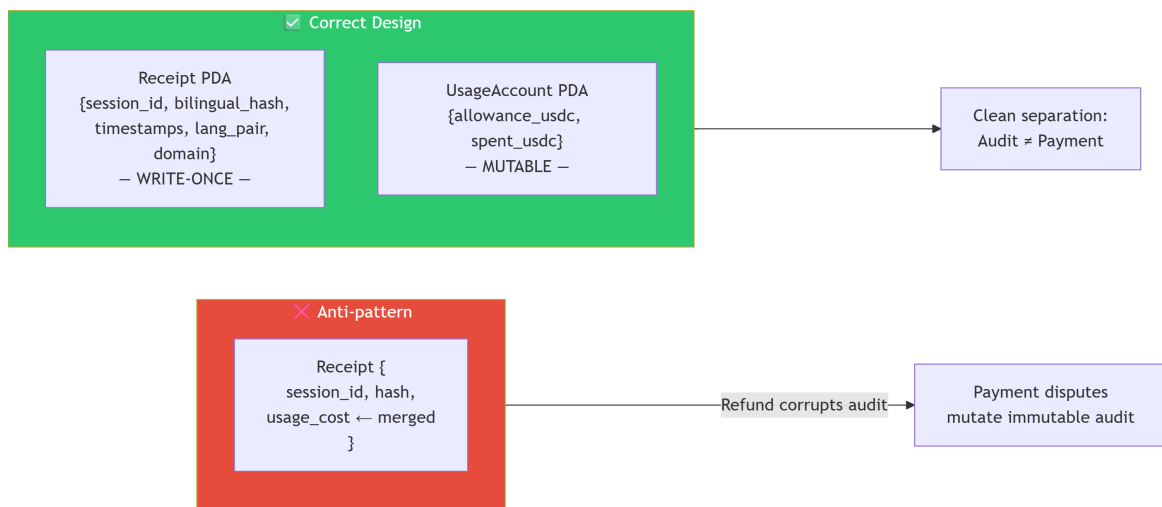
6.1 Finality Timeline

Solana Confirmed Finality

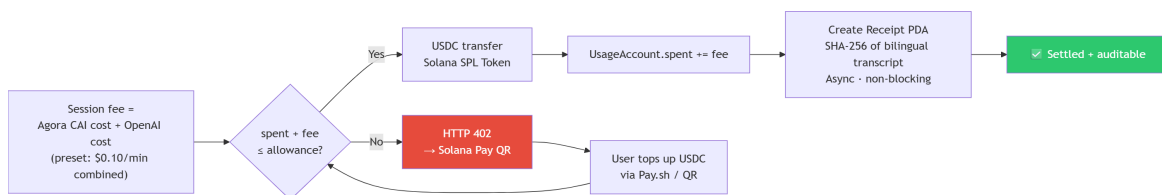


Receipts are async by design — no code changes needed when Alpenglw ships to mainnet.

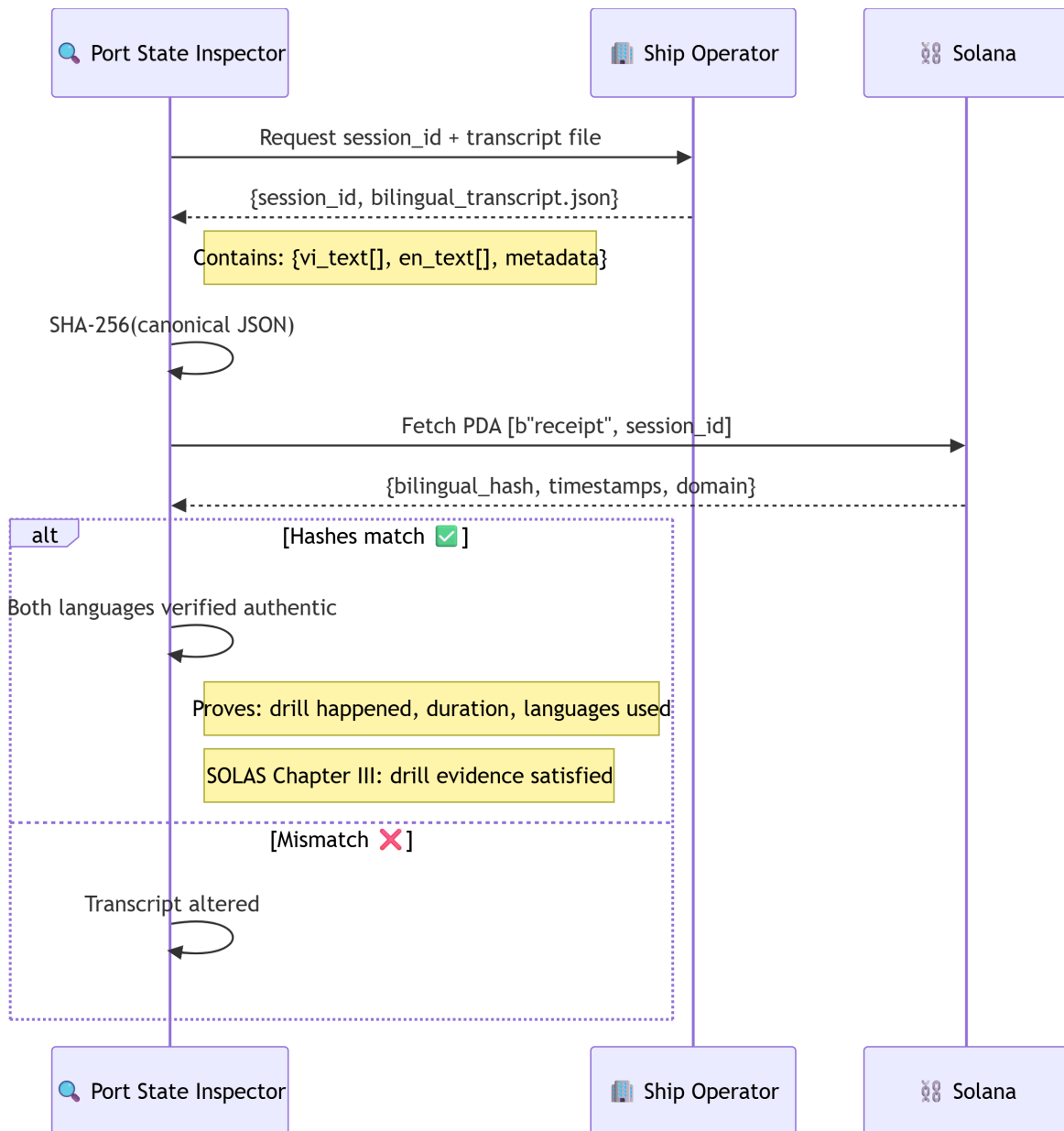
6.2 Two-Account Architecture



6.3 Payment Flow — x402 Pattern



6.4 SOLAS Audit Flow (Strengthened with Bilingual Hash)



7. Hardware Integration

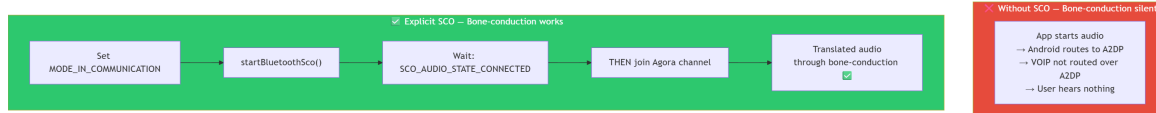
7.1 Hackathon Hardware — Shokz for Demo Day

Use Case	Model	IP	BT Codec	Notes
Port/ship worker	Shokz OpenRun Pro 2	IP55	SBC only (50-100ms)	Loudest bone-conduction; AI mic NC
Supervisor (demo)	Phone speaker	—	—	Sufficient for demo; judges hear EN output

Use Case	Model	IP	BT Codec	Notes
Aquatic coaching	Shokz OpenSwim Pro	IP68	SBC	BT above water only

Hardware warning: The OpenRun Pro 2 supports SBC codec only. No aptX, no aptX-LL. Plan for 50-100ms Bluetooth latency, not 20-40ms.

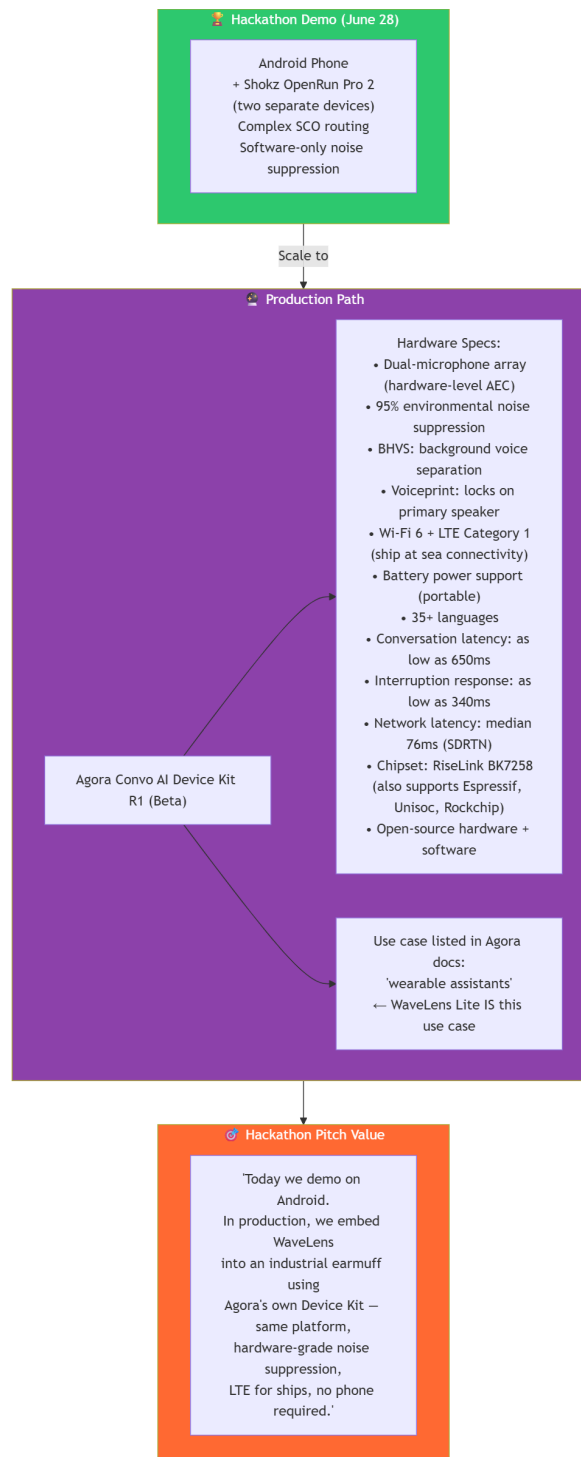
7.2 Android SCO Routing — Mandatory for Any Bone-Conduction Headset



Agora SDK with SCO: Use `AUDIO_PROFILE_SPEECH_STANDARD` + `AUDIO_SCENARIO_DEFAULT`.
Never use `MUSIC_HIGH_QUALITY` — conflicts with SCO.

7.3 Production Hardware Path — Convo AI Device Kit R1 (NEW)

This is the most significant find from reading Agora's live docs. It completely changes the production hardware story for WaveLens.



Why Device Kit solves the Bluetooth problem permanently: The Device Kit has a dual-microphone array with HARDWARE-level AEC. There is no Bluetooth SCO routing problem — the device captures, processes, and outputs audio natively. The bone-conduction output becomes a speaker directly integrated in the wearable.

For the hackathon: You cannot build a Device Kit prototype in 9 days. But you should have a slide showing the production path with the Device Kit architecture. Judges who know Agora's product line will be impressed.

8. Latency Budget (Updated with Official Device Kit Specs)

8.1 Verified Performance Numbers (From Live Agora Docs)

Metric	Hackathon (Android + Shokz)	Production (Device Kit)	Source
Conversation latency	~670ms P50	As low as 650ms	Agora Device Kit docs
Interruption response	~340ms	As low as 340ms	Agora Device Kit docs
SDRTN network latency	80-120ms	Median 76ms	Agora Device Kit docs
Noise suppression	Software, ~80dB	Hardware, 95% of noise	Agora Device Kit docs
Packet loss tolerance	80%	80%	Both confirmed
BT codec latency	SBC: 50-100ms	None (integrated)	Hardware

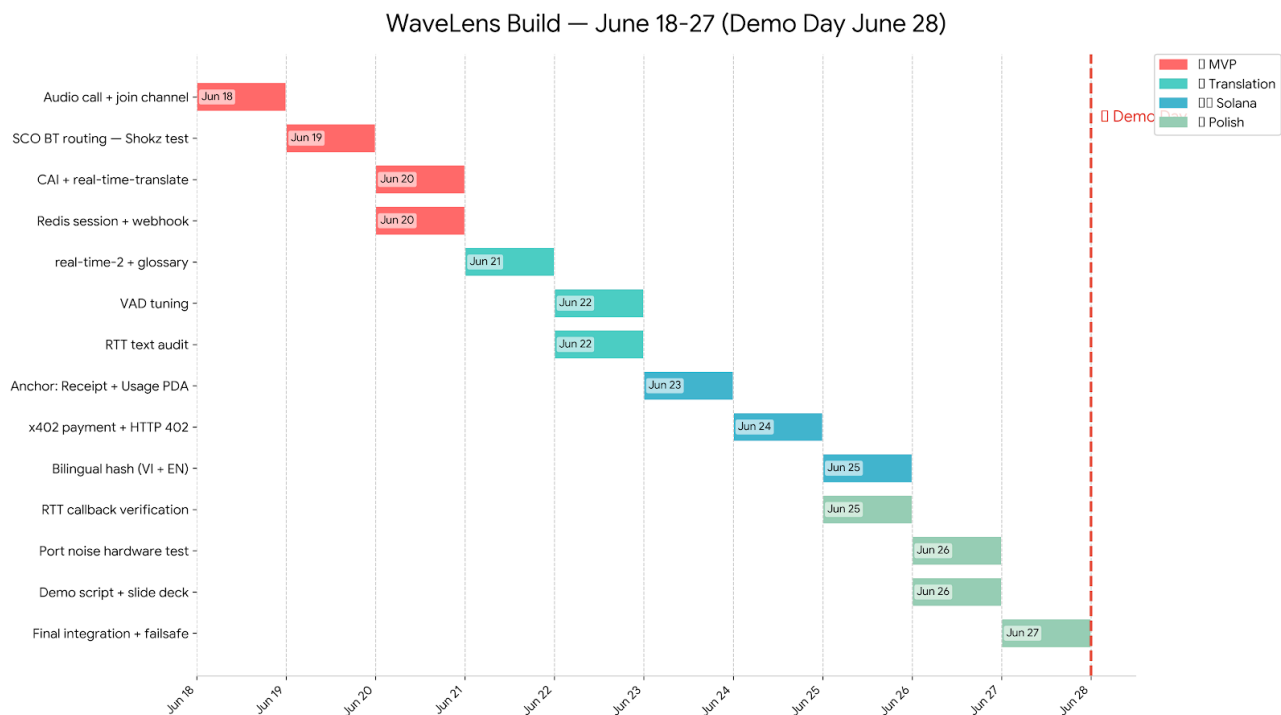
8.2 Latency Breakdown — Hackathon Android Path

Pipeline Step	P50 (Coaching)	P50 (Maritime)	P95 (100dB Noise)
Agora SDRTN + VAD	120ms	120ms	200ms
Model Inference	280ms (Translation)	380ms (Transl. + Reason)	600ms
Audio Delivery	80ms	80ms	150ms
Bluetooth SBC	75ms	90ms	150ms
Total E2E Latency	555ms	670ms	1,100ms

Target: "Under 1.5 seconds for 90% of turns" — achievable in normal conditions.

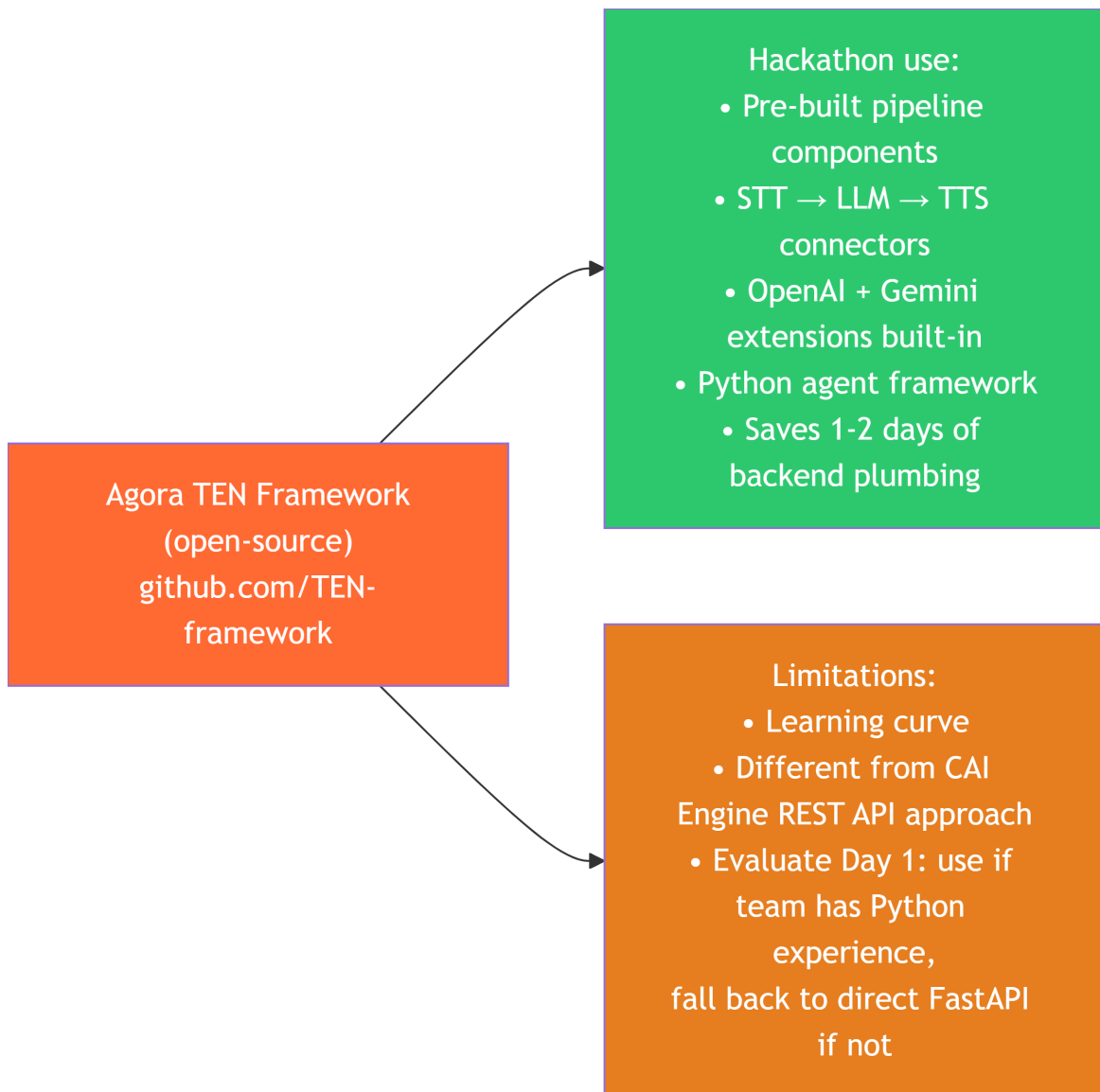
9. Hackathon Build Plan — 9 Days

9.1 Priority Timeline

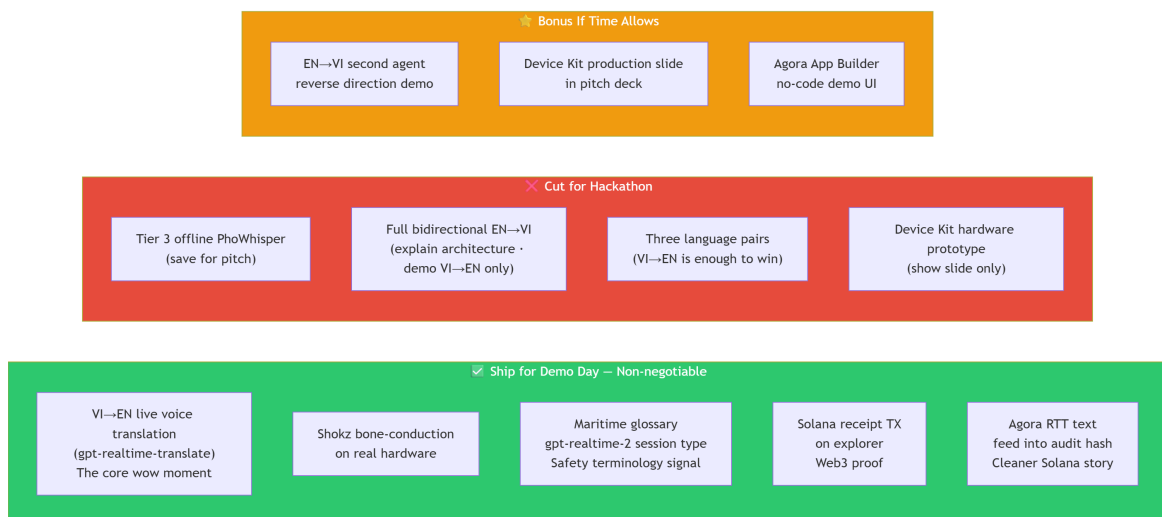


9.2 TEN Framework — Hackathon Accelerator (NEW)

Agora's TEN Framework (open-source, 6,000 GitHub stars) is a multimodal conversational AI pipeline framework. It provides pre-built components for STT → LLM → TTS pipelines. For the hackathon backend:



9.3 Cut vs Keep for 9-Day Build



9.4 Demo Day Narrative (June 28, ĐH Bách Khoa Đà Nẵng)

Opening hook (30 seconds):

"Da Nang Port is targeting 16.77 million tonnes in 2026. On every shift, a Vietnamese crane operator and a Korean ship officer work alongside each other through hand signals and guesswork. Walkie-talkies don't translate. Phones need hands. In-ear headphones block safety alarms. We built WaveLens Lite: voice-to-voice translation through bone conduction — so the operator hears the translation without blocking their ears, and both hands stay on the crane."

Live demo (4 minutes):

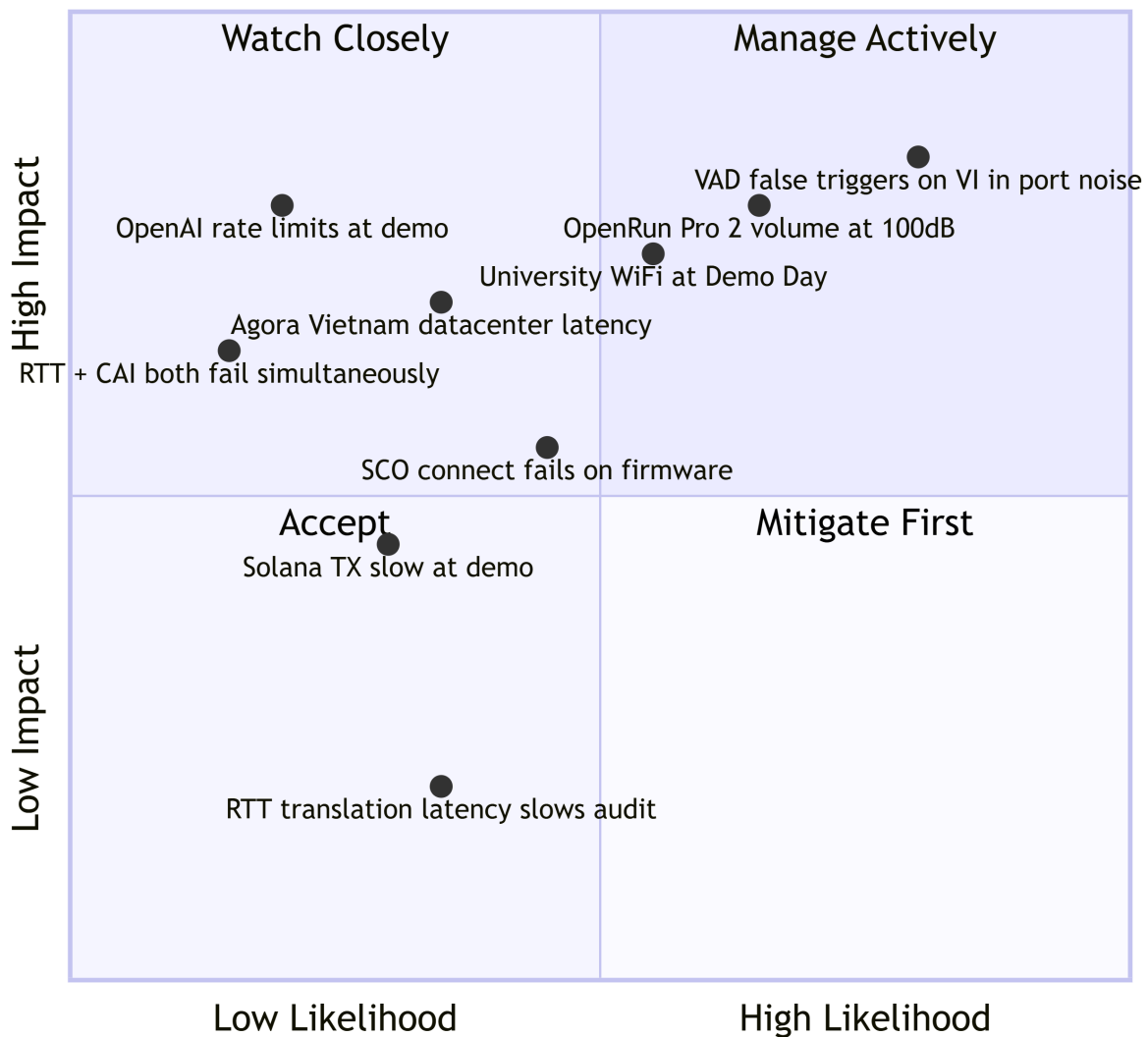
1. Show Shokz OpenRun Pro 2 — explain bone conduction in one sentence
2. Select "Maritime" domain → glossary loads (show the term list briefly)
3. Speak Vietnamese → English plays through phone speaker for the "supervisor"
4. Say "két ballast đang rò rỉ" → show "ballast tank is leaking" in the transcript UI
5. Open Solana explorer → show the receipt TX with the bilingual hash
6. Show Agora dashboard → session agent activity

Three judge questions answered before they ask:

- "Why Agora?" → 80% packet loss tolerance. Without it, crane yards with 40% radio congestion make translation fail. Raw WebRTC cannot do this.
- "Why Solana?" → SOLAS maritime law requires evidence of safety drills. An immutable on-chain hash of the bilingual transcript gives port state inspectors verifiable proof without storing any crew PII.
- "What's the production path?" → Agora already ships the Device Kit R1 for exactly this: a wearable with hardware-grade noise suppression, LTE for ships, and Agora's full AI pipeline embedded. Today we demo on Android; production deploys on the Device Kit.

10. Risk Assessment

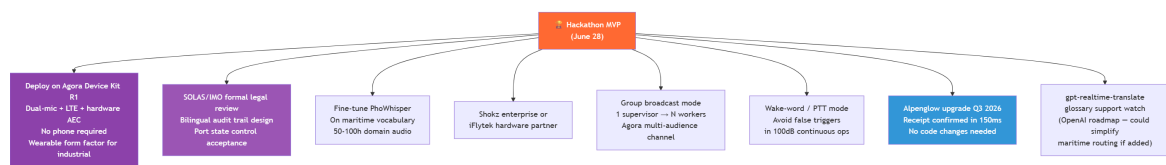
Risk Matrix – Likelihood vs Impact



Risk	Mitigation
University WiFi failure	Bring 4G/5G hotspot. Non-negotiable.
VAD false triggers on VI in industrial noise	Tune threshold=0.75, silence=800ms. Test on recorded port audio.
OpenRun Pro 2 too quiet at 100dB	OpenRun Pro 2 is loudest in class. Test in real loud environment before June 28.
OpenAI rate limits at demo	Pre-warm session before judges arrive. Gemini Live configured as hot fallback in Agora v2.6.
SCO connect fails	Test on the exact Shokz unit used for demo. Keep wired earphone as emergency audio proof.

Risk	Mitigation
RTT text lags (3s)	RTT runs async — receipt hash is computed post-session. No user-facing impact during demo.
Solana TX slow	Receipts are async. Add priority fee for demo sessions. Show "pending" state as expected.

11. Production Roadmap



Appendix A: Library Versions (June 2026 — Live Docs Verified)

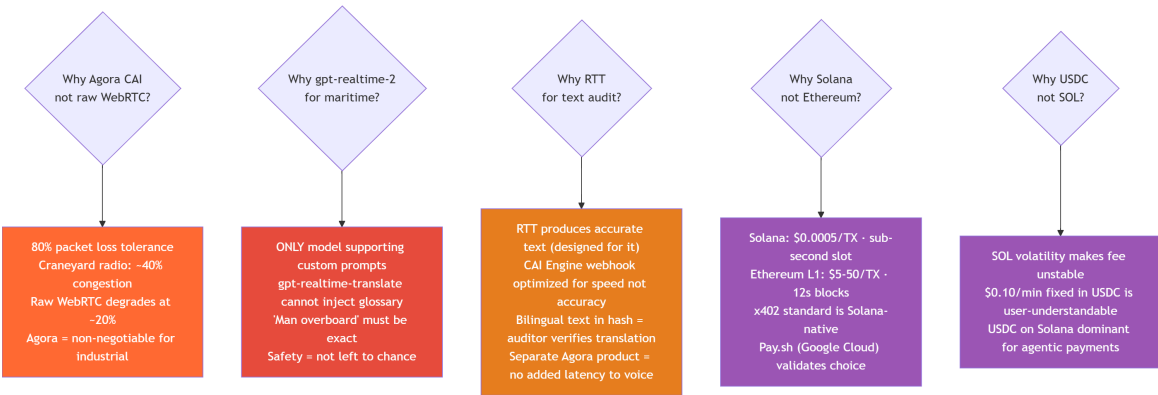
Library	Version	Critical Note
Agora Voice SDK (Android)	4.5.x	Confirm latest at docs.agora.io/en/sdks
Agora CAI REST API	v2 (engine v2.6)	Path: <code>/api/conversational-ai-agent/v2/</code>
Agora RTT API	v7.x	Path: <code>/api/speech-to-text/v1/</code>
OpenAI — maritime	<code>gpt-realtime-2</code>	Supports system prompt + glossary
OpenAI — coaching	<code>gpt-realtime-translate</code>	No custom prompts; \$0.034/min
Agora input transcription	<code>gpt-4o-transcribe</code>	Use language hint "vi" for Vietnamese
Anchor (Solana)	0.31.x	Use AVM (Anchor Version Manager)
Solana Web3.js	2.x	Breaking from 1.x — always use 2.x

Library	Version	Critical Note
@solana/spl-token	0.4.x	USDC SPL token transfers
PhoWhisper	medium	github.com/VinAIResearch/PhoWhisper — Tier 3 only
FastAPI	0.115.x	Async webhook handlers
Redis (aioredis)	2.x	Session state, 24h TTL

Appendix B: Maritime Glossary (Priority Order)

Vietnamese	English	Korean	Chinese	Priority
người rơi xuống biển	man overboard	남자가 떨어짐	人员落水	● CRITICAL
Bộ luật ISPS	ISPS code	ISPS 코드	ISPS规则	● CRITICAL
két ballast	ballast tank	밸러스트 탱크	压载水舱	● HIGH
bơm bi-lơ	bilge pump	빌지 펌프	污水泵	● HIGH
cần cầu cổng	gantry crane	갠트리 크레인	门式起重机	● HIGH
trạm tập hợp	muster station	집합소	集合站	● HIGH
xả đáy	blow-down	블로우다운	排污	● MED

Appendix C: Why Each Technical Choice



Appendix D: Agora Cost Model

Component	Free Quota	Paid Rate	\$600 Credits =
CAI Engine (Agora preset key)	300 min/month	\$0.10/min	6,000 minutes
Real-Time STT + Translation	300 min/month (shared with CAI)	~\$0.01/min (STT) + LLM cost	Large buffer
Agora RTC audio	10,000 min/month	\$0.99/1,000 min	Essentially free

For the hackathon: At 10 minutes per test session, the 300 free CAI minutes = 30 test sessions before credits activate. \$600 credits give 6,000 more minutes. A team doing 3 sessions/day for 9 days uses 270 minutes — well within the free tier alone.
